Hazardous Waste and Home Values: An Analysis of Treatment and Disposal Sites in the United States

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Abstract: There are 2,389 treatment, storage, and disposal facilities (TSDFs) of hazardous waste in the contiguous United States. We frame two sets of difference-in-differences and triple differences quasi-experiments that exploit the spatial and temporal variation in contamination and cleanup events and estimate hedonic price regressions using a sample of 9.7 million home transactions from 2000–2018. The discovery of contamination and subsequent investigation is associated with up to a 5% depreciation in the value of homes within 750 meters of a TSDF, but the evidence is mixed. In contrast, we find strong evidence that the completion of cleanup leads to an average 6%–7% appreciation. A total capitalization of $295 million can be attributed to the 195 TSDFs that were remediated since the inception of the Resource Conservation and Recovery Act. We estimate that the completion of cleanup yields an average lower-bound, ex post benefit of about $14,000 per household.

JEL Codes: D62, Q51, Q53

Keywords: cleanup, difference-in-difference, hazardous waste, hedonic, nonmarket valuation, property value, Resource Conservation and Recovery Act, RCRA

THE RESOURCE CONSERVATION AND RECOVERY ACT (RCRA) is a cornerstone of environmental policy in the United States. It regulates the generation, use, transportation, and eventual disposal of hazardous chemicals. The 1984 amendments to

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Dataverse data: https://doi.org/10.7910/DVN/9CP4M4

Received August 3, 2021; Accepted June 1, 2023; Published online January 25, 2024.

Journal of the Association of Environmental and Resource Economists, volume 11, number 2, March 2024.

https://doi.org/10.1086/726157

RCRA gave the US Environmental Protection Agency (EPA) the authority to establish requirements that facilities investigate, and if necessary, clean up, any hazardous releases into the environment. Such investigations and cleanup activities are collectively referred to as a corrective action. Releases of hazardous chemicals are truly ubiquitous, with over 35 million people (12% of the US population) living within one mile of a RCRA corrective action site (US EPA 2013). Cleaning up and reducing exposure to these chemicals (including arsenic, benzene, lead, mercury, and many others) reduces risks to human health and the environment.

Even though RCRA and the corrective action program have been in place for over three and a half decades, there is little known about the monetized benefits of the program. Many benefit endpoints of EPA regulations under the authority of RCRA go unmonetized in benefit-cost analyses (BCAs). We reviewed the BCAs for all major EPA regulations under the authority of RCRA over the last 10 years. These regulations affect when and to what extent investigations and cleanups are required. The only monetized benefits primarily entailed avoided regulatory or cleanup costs to regulators and industry. Welfare impacts to surrounding residents often go unmonetized and are represented solely in a qualitative fashion (see app. A for details; apps. A–G are available online).

The objective of our nationwide study is to estimate the monetized benefits of RCRA programs. Focusing on all hazardous waste treatment, storage, and disposal facilities (TSDFs) across the contiguous United States, we estimate a series of hedonic property value models. We set out to answer four main questions. First, is proximity to a TSDF associated with lower home values? Second, does the discovery of hazardous contamination and opening of a corrective action (CA) investigation further decrease nearby home values? Third, once contamination has occurred, does the subsequent cleanup and completion of a CA lead to an increase in values? Fourth, what are the welfare implications to households? We adapt an approach proposed by Banzhaf

Associated Universities (ORAU) under DOE contract number DE-SC0014664. Property transaction data were provided by Zillow through the Zillow Transaction and Assessment Dataset (ZTRAX). More information on accessing the data can be found at http://www.zillow.com/ztrax. Additional data support was provided by the Private-Land Conservation Evidence System (PLACES) at Boston University. We thank Shelby Sundquist for excellent research assistance. We are grateful to the EPA's Office of Resource Conservation and Recovery, and especially to Norman Birchfield, David Hockey, Joseph Krahe, Jennifer McLeod, and Scott Palmer for sharing their insights about the RCRA program and data. We thank Spencer Banzhaf, Heather Klemick, Lala Ma, and seminar participants at San Diego State University, the University of Calgary, the 2020 and 2021 PLACES webinars, and the Northeastern Agricultural and Resource Economics Association's 2021 virtual conference, for helpful comments on earlier versions of this research. We also thank Sheila Olmstead and three anonymous referees for helpful feedback. The results and opinions are those of the authors and do not reflect the position or policies of the US EPA, DOE, ORAU/ORISE, or the Zillow Group. Any errors are those of the authors.

(2021) to obtain lower-bound estimates of the ex post benefits of cleanup to impacted households. This is the first nationwide analysis to use high-resolution transaction-level data to assess the impacts of the RCRA program and, to our knowledge, is among the first to apply Banzhaf's framework to obtain bounding welfare estimates.

We frame two sets of spatial difference-in-differences (DID) and triple differences quasi-experiments. The first quasi-experiment examines the impacts of the opening of a CA, compared to the pre-CA period. The opening of a CA may reflect new contamination and/or signal new information that changes households' risk perceptions. Increases in perceived risks may in turn result in a decrease in nearby home values. That said, it could also be that a new CA, on average, does not present new information and pollution concerns. CAs often involve investigations of historical contamination issues (US EPA 2013). Any perceived risks may already be capitalized in surrounding home values.

Of primary interest is the second quasi-experiment, which examines how, after contamination is present and the CA is opened, property values respond to completion of cleanup and closure of the CA. Closure of a CA and the preceding visual cues associated with cleanup may signal to residents that the pollution issue is resolved and, thus, lead to an increase in nearby home values.

Both quasi-experiments exploit temporal and spatial variation in contamination and cleanup events and are examined within the same hedonic price models, but the corresponding pretreatment periods differ. In subsequent models we use coarsened exact matching (CEM) to make the treatment and control groups more comparable and, ultimately, to provide more defensible causal estimates.

A national data set of 2,389 TSDFs, and details about the CA and cleanup activities that occurred at 689 of these facilities, is compiled from RCRAInfo (an extensive database maintained by the EPA). These data are spatially and temporally linked to residential property transactions data from Zillow (2020) and supplemented with high resolution spatial data of residential parcels and nearby amenities compiled by the Private-Land Conservation Evidence System (PLACES) at Boston University. Our data set covers most of the contiguous United States and entails 9.7 million home transactions from 2000 to 2018.

We find that proximity to a TSDF is associated with lower home values. This negative association is strongest nearest a TSDF (i.e., within 250 meters)—suggesting an almost 8% lower value, all else constant. Although this negative association diminishes with distance, it remains significant out to 4.75 kilometers. Our base models suggest an additional 5% decline in the value of homes within 750 m of a TSDF when a CA is opened, but the evidence is mixed, and diagnostics of the pretreatment trends suggest that a causal interpretation is questionable.

In contrast, we find strong evidence that once cleanup is complete and the CA is resolved, home values within 750 m increase by 6%–7%, on average. Data diagnostics support a causal interpretation of this post-CA appreciation. This translates to an average increase in value of about $11,600 (2019$) per home. Extrapolating this result to all single-family homes within 750 m of one of the 195 TSDFs that have been cleaned up since the inception of the RCRA CA program suggests a total increase in housing stock value of $295 million. Finally, among residents living within 750 m of a CA site, we estimate that the completion of cleanup leads to an average lower-bound, ex post benefit of $14,000 per household.

Next, we briefly review the relevant hedonic property value literature on proximity to contaminated sites and cleanup. In section 2, we provide details on the RCRA CA program and data. Section 3 describes the empirical methods, and the results are presented in section 4. Section 5 outlines the approach for estimating a lower bound of the ex post welfare effects and presents the results. Concluding remarks, policy implications, and directions for future research are discussed in section 6.

1. LITERATURE

Kohlhase (1991) and Michaels and Smith (1990) were among the first to study how proximity to waste sites impacts home values. The ensuing literature on the topic is large and well established, with several literature reviews conducted (Farber 1998; Boyle and Kiel 2001; Jackson 2001). A meta-analysis by Schütt (2021) revisited the literature and synthesized the results across 83 hedonic studies. Schütt concludes that the literature generally suggests that proximity to waste sites does lower home values and that this adverse effect is strongest among the most severe and hazardous sites. Schütt also finds that cleanup of the contamination reduces these adverse price effects.

The rise of quasi-experimental methods in environmental economics (Greenstone and Gayer 2009), and in particular in hedonic property value applications (Parmeter and Pope 2013), has led several to reexamine the causality of the empirical relationship between home values and proximity to contaminated sites. Such efforts include a series of national-scale studies on various cleanup programs headed by the EPA's Office of Land and Emergency Management (OLEM). Haninger et al. (2017) analyze how remediation through EPA's brownfield cleanup grants program impacts home values and find that cleanup leads to a 5%–15% appreciation among homes within 2 km. Focusing on the most severe and well-publicized releases of petroleum contamination from underground storage tanks at gas stations, Guignet et al. (2018) estimate site-specific DID hedonic price equations and then synthesize the results in an internal meta-analysis. They find an average depreciation of 6% in value among homes within 3 km of a release, and they also find that home values fully rebound, on average, after contamination is cleaned up. Gamper-Rabindran and Timmins (2013) use decadal census tract data to examine the impact of Superfund site cleanups on home values. Superfund sites are those that fall under the jurisdiction of, and are subsequently cleaned up under, the Comprehensive Environmental Response, Compensation and Liability Act (CERCLA) and are among the most notorious and severe hazardous waste sites in the United States. Gamper-Rabindran and Timmins find that homes within 3 miles (about 4.8 km) appreciate by roughly 20% when a Superfund site is cleaned up and that this effect is strongest among the lowest quantiles of the tract-specific price distributions. Although hazardous waste contamination is remediated through both RCRA and CERCLA, the sites targeted are often different, and so transferring Gamper-Rabindran and Timmins's results to our current context is not necessarily appropriate. CERCLA generally deals with remediation at nonoperating, often abandoned sites. In contrast, RCRA is meant to manage solid and hazardous waste at facilities currently in operation and where the owners and operators are known.

Cassidy et al. (2022) examine how cleanups under RCRA's CA program impact home values. Using decadal census tract-level data and a methodology similar to Gamper-Rabindran and Timmins (2013), they find that, after cleanup, home values in the same tract as the RCRA site increase in value by up to 11%. They also find that these price effects are strongest among homes in the lowest portions of the tract-specific price distributions.

Our study and Cassidy et al.'s (2022) are useful complements. Cassidy et al. focus on a more comprehensive set of RCRA sites, including large quantity generators, recycling facilities, and TSDFs. Our analysis focuses on just TSDFs, the highest risk and most hazardous category of RCRA sites. That said, our analysis and use of higher-resolution, revealed-preference data yields several advantages. Cassidy et al. use temporally and geographically coarse decadal census tract-level data, which hinders identification of very local and possibly dynamic impacts on home values. Another potential drawback is that the census data are based on self-reported home values.[1] Our analysis uses actual prices from individual residential property transactions. We know the location of each home and the date of the transaction. This finer temporal and spatial resolution allows us to identify very local effects, to better isolate the price effects of interest from spatially correlated omitted variables, and to examine dynamic effects in how both the opening and subsequent cleanup and completion of the CA impact home values. In addition, we utilize data on all 2,389 TSDFs in the contiguous United States and not just those where a CA occurs. This provides an additional layer of controls, which we exploit in our triple differences models.

2. BACKGROUND AND DATA

We compile and combine two detailed and comprehensive nationwide data sets—one of all RCRA sites and corrective actions (CAs), and the other of most residential properties and transactions in the United States from 2000 through 2018.

1. Kiel and Zabel (1999) found that self-reported home values tend to be 5.1% higher than actual values, on average. Although to be fair, Kiel and Zabel conclude that such differences are not related to house or neighborhood characteristics and that regression models based on self-reported values can yield reliable implicit price estimates of house and location attributes.

2.1. TSDFs and Corrective Actions

A hazardous waste treatment, storage, and disposal facility (TSDF) is a permanent establishment in operation for the purpose of waste management (US EPA 2005). Examples of TSDFs include active landfills, lead battery recycling plants, storage facilities for cleaning by-products, and steel mills and plumbing manufacturers that treat or dispose of hazardous waste on-site. We focus on TSDFs because they are the riskiest and most hazardous type of RCRA sites. About 43% of all CAs in our data involve a facility that is registered as a TSDF. Almost 30% of all TSDFs are involved in a CA. TSDFs face increased requirements to prevent releases and exposures (US EPA 2005) and are among the highest priority for EPA's Office of Resource Conservation and Recovery (ORCR).

A data set of all RCRA-regulated facilities was first compiled from RCRAInfo—an extensive multi-relational database compiled and maintained by the EPA.[2] Geographic coordinates for the RCRA sites were obtained from EPA's Federal Registry Service (FRS) database.[3] In the end, a data set of 129,670 RCRA sites across the entire United States was compiled. This includes all sites that were confirmed small or large quantity generators, recyclers, and/or TSDFs for at least one federally regulated hazardous material. CAs involving contamination releases were identified at 1,610 of these sites.[4]

We focus on the 2,389 facilities that are designated as a TSDF in the data and that are located in the contiguous United States. CAs occurred at 689 of these TSDF sites (see fig. 1). The 1,700 TSDFs where a CA involving actual contamination did not occur serve as a control group. At 195 of the 689 CA sites, the CA was completed by the end of 2018. This means that the site-specific remediation and/or exposure mitigation goals were achieved or were sufficiently under way so that the contamination was no longer deemed a threat.

The RCRAInfo data include several different types of events and milestones associated with the CA process. Although the database is standardized, event dates are sometimes missing, or certain events are not applicable or always inputted in a consistent fashion. After numerous discussions with EPA's ORCR, we define the opening of a corrective action as the earliest of the following three events: when regulators were first notified of contamination, the date an investigation was imposed, or when the initial assessment was complete.[5] We focus only on CAs where the investigation revealed actual contamination that required cleanup or active efforts to minimize exposure, as described below.

2. EPA, RCRAInfo Public Extract, https://rcrapublic.epa.gov/rcra-public-export/ (data downloaded in June 2019).

3. EPA, FRS Data Resources, https://www.epa.gov/frs/frs-data-resources (data downloaded in August 2019).

4. See app. B for additional details.

5. This corresponds to the RCRAInfo event codes CA060, CA100, and CA200, respectively.

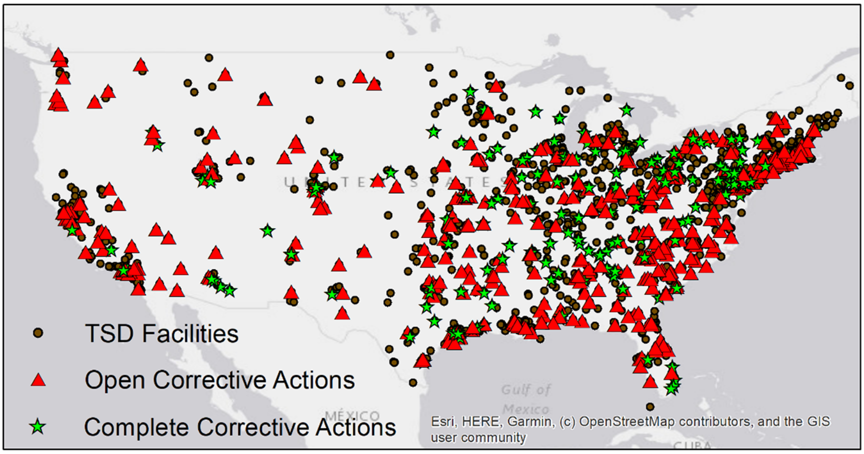

Figure 1. Treatment, storage, and disposal facilities (TSDF) and corrective actions

As seen in table 1, almost all CA sites of interest (98%) are also on the EPA and state regulators' list of high priority RCRA cleanups (US EPA 2019). The average CA has a risk prioritization between "medium" and "high." By definition, all CAs of interest involved one or more of the following three approaches to address the hazardous waste contamination. In 76% of the cases, active remediation technologies are

Table 1. Descriptive Statistics of Corrective Actions (CAs)

Variable[a]	Mean	SD	Min	Max
Identified as high priority by ORCR	.975	.155	0	1
Risk prioritization (1 = high, 2 = medium, 3 = low)[b]	1.640	.791	1	3
Risk prioritization missing	.028	.164	0	1
Remedy constructed	.763	.425	0	1
Physical controls implemented	.409	.492	0	1
Institutional controls implemented	.486	.500	0	1
CA complete for entire facility	.283	.451	0	1
CA duration (years)[c]	15.09	9.12	.00	30.94

Note. Unless otherwise noted, descriptive statistics are for all n = 689 unique TSDFs in the contiguous United States where a corrective action (CA) involving actual contamination was opened.

[a] Variables are all binary indicators unless otherwise noted.

[b] The risk prioritization variable has missing values for 19 of the 689 facilities. These missing values are excluded from the risk prioritization descriptive statistics.

[c] CA duration calculated for the 142 (out of 195) TSDFs where the CA was marked as complete in RCRAInfo by the end of 2018 and where the opening date was not missing. Duration was coded as missing for three sites where the CA opening date was listed as occurring after the completion date.

constructed and implemented (e.g., excavation of contaminated soil, pump-and-treat groundwater). Physical and engineering controls to minimize exposure and migration of contaminants are employed at 41% of the CA sites. Such controls include routine monitoring and testing of groundwater, concrete caps, and physical barriers. Institutional controls (e.g., zoning restrictions on future land uses) are put in place at 49% of the CA sites.[6] It is not uncommon to have two (172 CAs) or even all three (141 CAs) of the approaches in place to remediate contamination and minimize exposure.

At 195 of the 689 CAs (28%), the investigation and cleanup was complete by 2018 (the end of our study period). We define the completion of a CA as the earliest recorded date for one of the following events and only when applicable to the entire site—when the CA was officially terminated or when the required performance standards were achieved.[7] Although the cleanup goals have been achieved and active cleanup efforts have ceased by this point, some institutional or physical controls, as well as routine monitoring, may remain. The average (and median) duration of a CA is 15 years, but there is notable variation, ranging from less than one to almost 31 years.

The timing of when these investigations and cleanup activities are initiated and subsequently resolved varies across the sites (see fig. A1; figs. A1–A9 are available online). This temporal variation, coupled with the spatial variation in the location of these CA sites, bolsters the quasi-experimental design discussed in section 3. Given the large number of sites spread across such a vast study area, with CAs being initiated and completed in different years, it is less plausible that potential confounders are systematically correlated with both the location and timing of contamination and cleanup events.

Although we do not observe what hazardous materials were released at each of the 689 CA sites, we do know what chemicals were registered for use (see fig. A2). The general categories of ignitable and corrosive waste are the most common, followed by spent nonhalogenated solvents (e.g., xylenes, acetone, ethyl benzene), which are associated with increased risks of neurological damage, nausea and headaches, kidney damage, and cancer (US HHS 1994, 2010; US EPA 2021). Lead is the most common individual chemical and is associated with numerous adverse health outcomes, including cancer, kidney, and cardiovascular problems, and even mortality in adults (US EPA 2013; US HHS 2020; Klemick et al. 2022). Lead is also well known to cause cognitive deficits in children (Lanphear et al. 2005; Miranda et al. 2007; Aizer et al. 2018; US HHS 2020), and evidence suggests that such deficits (Shadbegian et al.

6. The implementation of active remediation technologies, physical controls, and institutional controls is identified by the following event codes: CA550RC (remedy construction); CA770GW and CA770NG (groundwater and nongroundwater controls); and CA772EP, CA772GC, CA772ID, and CA772PR (institutional controls).

7. This corresponds to the RCRAInfo event codes CA999, CA999NF, CA999RM; and CA900CR and CA900NC, respectively. We thank Jennifer McLeod at EPA for providing data of when these events applied to the entire facility (personal communication, November 19, 2019).

2019) and other psychological effects (Schwaba et al. 2021) can persist for years. Other common heavy metals like cadmium and mercury are associated with an increased risk of kidney damage, and common contaminants like benzene, tetrachloroethylene, and arsenic are carcinogenic and associated with increased risks of cardiovascular issues (US EPA 2013, 2021).

Studies on similar hazardous waste sites where several of the aforementioned contaminants were present have shown that proximity to these sites adversely affects health and cognitive outcomes (Rau et al. 2015; Klemick et al. 2020; Persico et al. 2020). Cassidy et al. (2021) find that contamination of public water sources is at least one pathway for which residents are exposed to hazardous releases specifically from RCRA sites. After cleanup they find that the levels of hazardous chemicals are significantly reduced among public water systems where the source water is within 1 km of the RCRA site. They also find that incidents of low birth weight are significantly reduced by cleanup, particularly among black populations.

To examine whether actual and perceived health risks, and other factors like displeasing aesthetics and degraded environmental quality are capitalized into local home prices, we spatially and temporally link the TSDF and CA data to residential property transactions.

2.2. Residential Properties and Transactions

Information on residential parcels and transactions across most of the United States comes from Zillow's ZTRAX program (Zillow 2020). The ZTRAX data include information on transaction prices and dates, types of transactions, addresses, types of homes, and house structure characteristics. The geographic coordinates provided by ZTRAX are sometimes incomplete, and depending on the spatial scale of the environmental commodity of interest, these coordinates may not be sufficiently precise.[8] Given the localized nature of disamenities like RCRA sites, the ZTRAX data are supplemented with higher-precision digital parcel maps obtained from 12 open-source state-level data sets and by data from two commercial providers (Regrid and Boundary Solutions, Inc.). Details on these data and sources are provided by Nolte (2020).

The geographic data were processed using the Private-Land Conservation Evidence System (PLACES). PLACES uses assessor parcel numbers to link ZTRAX data to parcel boundaries based on county- and town-specific deductive string pattern matching and geographic quality controls (Nolte 2020). Within a geographic information system (GIS), residential parcels are spatially linked to the corresponding 2010 census tract. Distance to the nearest highway and the proportion of developed land

8. Geographic coordinates in the ZTRAX database have a range of known issues and errors, including different geographic datums across jurisdictions and missing and incorrect coordinates. Nolte et al. (forthcoming) provide a detailed assessment of these issues and describe ways in which PLACES improves the accuracy and completeness of these data.

within a 200 m and 500 m circular buffer around each home, are also calculated.[9] Most importantly, the Euclidean distance from each residential parcel to each TSDF within 10 km was calculated.

Our analysis focuses on all arms-length, single-family home transactions in the contiguous United States from 2000 to 2018 where the home was within 5 km of a TSDF. After eliminating outliers, the final sample size is $n = 9{,}763{,}582$ transactions.[10] The study area covers 837 counties across 47 states and Washington, DC.[11]

Descriptive statistics of the residential transactions are presented in table 2. The mean sales price is about $272,670 (2019$). The average home has a lot size of about 0.25 acres, 1.4 stories, 1.86 bathrooms, an interior square footage of 2,958 square feet, and is 46 years in age. There are some missing values for all of these house and parcel characteristics. As can be seen by the companion missing value dummies, data on acreage and interior square footage are only missing for 2.6% and 5.1% of the sample, respectively. House age is missing for about 6.7% of the sample. A larger proportion of the sample is missing data on the number of stories (19.1%) and bathrooms (23.0%). When applicable, observations with missing values are excluded from the corresponding statistics in table 2. In the later analysis, missing values are coded as zero and companion missing dummy variables are included as independent variables in the regression models.

Turning to the location variables in table 2, the proportion of land area that is developed within 200 m and 500 m of a parcel is 54% and 50%, on average, and about 39% of transactions are of homes within 500 m of a highway. On average, there are 0.03 TSDFs within 750 m of a home, and 0.14 and 1.73 within 1,500 m and 5,000 m, respectively. We later measure proximity to a TSDF using 250 m incremental bins. Given the large scale of the analysis, even among the smallest bins nearest to a TSDF there are several thousand sales. For example, there are 19,059 home sales within 0–250 m of a TSDF,

9. Census tract data come from the National Historical Geographic Information System (Manson et al. 2018). Land cover data come from the 2011 National Land Cover Database (Dewitz 2019). The highways data are from the US Census Bureau's TIGER/Line shapefiles (2019).

10. Outlying observations were dropped as follows: homes with a real price (2019$) less than $20K or more than $1M (corresponds roughly to the 5th and 95th percentiles), less than 0.05 acres or greater than 2 acres (corresponds roughly to the 5th and 95th percentiles), less than one or greater than three stories, an interior size of less than 750 square feet or greater than 12,000 square feet (corresponds roughly to the lowest and highest percentiles), less than one or greater than six bathrooms, and if the home was greater than 150 years in age.

11. The ZTRAX data do include information on some transactions in nondisclosure states where housing transaction data are often not as widely available, including Idaho, Kansas, Louisiana, Mississippi, Montana, New Mexico, North Dakota, Texas, Utah, and Wyoming (Wentland et al. 2020). We maintain these transactions in our main analysis, but the results are robust to the exclusion of these nondisclosure states.

Table 2. Residential Transactions Descriptive Statistics

Variable	Observations	Mean	SD	Min	Max
Price (2019$)	9,763,582	272,669	202,034	20,001	999,982
Acres	9,510,481	.252	.236	.050	2.000
Missing: acres†	9,763,582	.026	.159	0	1
Stories	7,897,251	1.409	.503	1.0	3.0
Missing: stories†	9,763,582	.191	.393	0	1
Bathrooms	7,518,058	1.857	.760	1	6
Missing: bathrooms†	9,763,582	.230	.421	0	1
Interior square footage	9,267,429	2,958	1,906	750	12,000
Missing: interior square footage†	9,763,582	.051	.220	0	1
Age (years)	9,105,291	45.980	29.519	0	150
Missing: age (years)†	9,763,582	.067	.251	0	1
% land developed within 0–200 m	9,763,582	54.445	28.461	0	100
% land developed within 0–500 m	9,763,582	50.308	23.536	0	100
Highway within 500 m†	9,763,582	.388	.487	0	1
No. of TSDFs within 0–750 m	9,763,582	.03	.17	0	3
No. of TSDFs within 0–1,500 m	9,763,582	.14	.40	0	6
No. of TSDFs within 0–5,000 m	9,763,582	1.73	1.28	1	18

Note. The final sample includes $n = 9{,}763{,}582$ single-family home transactions. Descriptive statistics for some variables are for a smaller sample due to missing values, as reflected by the corresponding missing value indicators.

† Variables denoted with † are binary indicators.

and 84,630 within 250–500 m. Beyond that there are well over 100,000 transactions in each 250 m bin (see fig. A3).

There is also a reasonable number of observations in many of the 250 m distance bins during each of the three CA stages—pre–corrective action, mid–corrective action, and post–corrective action (see fig. A4).[12] This is particularly true for the mid-CA stage, which is not surprising because the average CA is opened for 15 years. Even in the nearest distance bins there are often at least a few hundred transactions observed. The relatively small sample size in some of the nearest distance-by-stage bins illustrates the benefit of conducting such a large nationwide study. Using a large-scale data set allows us to obtain a reasonable number of identifying observations for statistical analysis, while still maintaining a flexible specification of the distance gradient. Such a flexible and detailed price gradient specification would not be possible in a more local analysis where fewer transactions are observed.

12. Although it is possible for a house to have a nearby TSDF in one stage of the CA process and simultaneously another nearby TSDF in a different stage, this is uncommon. We observe only two such transactions in the data.

3. METHODS

3.1. Difference-in-Differences and Triple Differences Framework

The usefulness of the spatial DID strategy for causal inference in the hedonic pricing framework was made clear through early applications by Davis (2004) and Linden and Rockoff (2008), among others. There have since been numerous hedonic applications of the DID approach to estimate the impact of local environmental commodities on surrounding home values, including studies of contaminated site cleanups (Haninger et al. 2017; Guignet et al. 2018), aquatic vegetation and invasive species (Horsch and Lewis 2009; Guignet et al. 2017), flood and hurricane risks (Atreya et al. 2013; Bin and Landry 2013), and shale gas extraction (Muehlenbachs et al. 2015). The approach remains at the forefront of hedonic pricing methods (Guignet and Lee 2021).

The intuitive nature of the DID and triple differences strategies in mimicking a classical experimental design make the approaches particularly attractive. Consider the simplified illustration in figure 2. The top row represents homes in the neighborhood around a TSDF where contamination and a CA will either eventually occur but has not yet (top left panel) or has occurred (top right panel). The bottom row represents homes near a counterfactual TSDF where contamination and a CA do not occur. The price effect of interest is experienced by homes in group *B*. These are homes

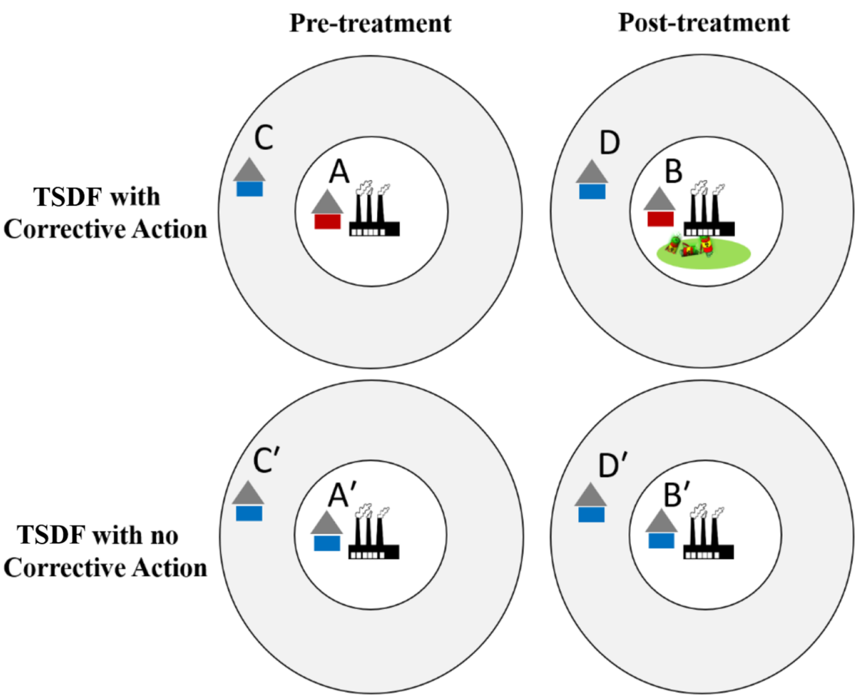

Figure 2. Spatial difference-in-differences illustration

in close proximity to a TSDF after a CA is initiated. In quasi-experimental terms, group B is the treated group, posttreatment. Our first objective is to identify how contamination and the opening of a CA impact home values. A simple first-difference estimate comparing before and after the opening of a CA would be $B - A$. However, home values in neighborhoods where contamination and CAs occur could be changing over time for other unobserved reasons, which would confound such a first-difference estimate. We can frame a spatial DID estimate that uses homes in the same community as the counterfactual but that are far enough away from the CA site so as not to be impacted. In doing so, broader neighborhood-specific trends are differenced out. Our spatial DID strategy is expressed as $(B - A) - (D - C)$.

Such DID estimates could be biased, however, if the trends in home values near versus far from a TSDF are systematically different for unobserved reasons. We frame a triple differences estimate that addresses that possibility by utilizing home sales around the control set of TSDFs where no contamination is discovered and CAs do not occur. Our triple differences estimate in this simple illustration is represented as $(B - A) - (B' - A') - \{(D - C) - (D' - C')\}$.

Our second objective is to identify how cleanup and the completion of a CA impact home values. A similar set of DID and triple differences comparisons is used to examine how the completion of a CA impacts home values, relative to homes near an open CA. Although we view these as two separate quasi-experiments, where we compare different treatment events relative to different counterfactuals and baseline periods, they are estimated within the same hedonic price models.

3.2. Hedonic Property Value Model

Our spatial DID and triple differences identification strategies are implemented within a hedonic regression framework, where the dependent variable is the natural log of the price of home i, in neighborhood j, in housing market m, and at time t (p_{ijmt}). More formally:

$$\begin{aligned}\ln\left(p_{ijmt}\right) = {} & \boldsymbol{x}_{ijmt}\boldsymbol{\beta}_{mt} + \mathbf{TSDF}_{ijm}\boldsymbol{\varphi} + \mathbf{pre}_{ijmt}\boldsymbol{\gamma}^{\text{pre}} + \mathbf{mid}_{ijmt}\boldsymbol{\gamma}^{\mathbf{mid}} \\ & + \mathbf{post}_{ijmt}\boldsymbol{\gamma}^{\text{post}} + \boldsymbol{\tau}_{mt} + \upsilon_{jm} + \varepsilon_{ijmt}.\end{aligned} \tag{1}$$

The right-hand-side control variables include a vector of house and neighborhood characteristics ($\boldsymbol{x}_{ijmt}$), which includes the number of stories and bathrooms, the natural log of the lot size (in acres) and interior square footage, measures of the percentage of the immediate area that is developed, and proximity to highways. As reflected by the subscript on the $\boldsymbol{\beta}_{mt}$ coefficient, the hedonic price surface with respect to these attributes is allowed to vary by housing market (i.e., county) and year. Although not explicitly represented for notational ease, we include county × year interaction terms with all elements of $\boldsymbol{x}_{ijmt}$. We are assuming a single nationwide hedonic price surface when estimating equation (1), but these interaction terms allow the equilibrium price surface to vary across markets and over time in these dimensions. Additionally, county-specific year and quarter fixed effects ($\boldsymbol{\tau}_{mt}$) are included to capture broader, market-specific trends and seasonal effects. The term v_{jm} denotes spatial fixed effects at the census tract level to account for all time-invariant factors associated with a specific location. Last, ε_{ijmt} is a normally distributed disturbance term, which we allow to be correlated for all transactions within the same county.

The parameters of direct interest are $\boldsymbol{\varphi}$, $\boldsymbol{\gamma}^{\text{pre}}$, $\boldsymbol{\gamma}^{\text{mid}}$, and $\boldsymbol{\gamma}^{\text{post}}$. The vector $\mathbf{TSDF}_{ijm}$ denotes the number of TSDFs within each 250 m incremental bin.[13] Estimates of $\boldsymbol{\varphi}$ are used to estimate the price-distance gradient associated with proximity to TSDFs, irrespective of a CA. The variable $\mathbf{pre}_{ijmt}$ is a vector of indicator variables denoting the presence of a TSDF in each of the incremental distance bins around a home where a CA will occur but has not yet occurred as of the time of the sale. Both $\mathbf{TSDF}_{ijm}$ and $\mathbf{pre}_{ijmt}$ are needed because homes may be near TSDFs where no contamination issues requiring a CA occur, but other disamenity effects associated with TSDFs could affect home values and may be important to control for due to the tendency for TSDFs to be spatially clustered.

The first treatment event vector is $\mathbf{mid}_{ijmt}$, which denotes the presence of an open CA within each distance bin around a home. The second treatment event vector is $\mathbf{post}_{ijmt}$, which denotes the presence of a completed CA in each bin. The coefficients $\boldsymbol{\gamma}^{\text{pre}}$, $\boldsymbol{\gamma}^{\text{mid}}$, and $\boldsymbol{\gamma}^{\text{post}}$ reflect the incremental difference in home values for a CA at each distance bin and in each respective CA stage, relative to a TSDF where a CA never occurs. In that sense, these parameters by themselves can be interpreted as cross-site first difference estimates and represent the first differencing in our triple differences framework. For the main regression models, as informed by initial data diagnostics (see sec. 3.3), a 0–750 m bin is assumed for the treated group and a broader 750–1,500 m bin for the control group.

Within this single regression model, we frame two separate quasi-experimental comparisons. The first is of how home values are impacted by the discovery of contamination and opening of a CA, relative to before the CA was opened. The second focuses on how the subsequent cleanup and completion of the CA affects home values, relative to homes around an opened CA (i.e., in the mid-CA stage). These two quasi-experimental comparisons are relative to two different baselines, as can be seen in the

13. Notice that there is no t subscript on $\mathbf{TSDF}_{ijm}$. We assume that the number of TSDFs in each distance bin is constant during our study period. Unfortunately, data on when a TSDF began its operation are sparsely populated and are only available for 330 of the 2,389 TSDFs in our study (13.8%). Among the 330 TSDFs where a beginning date is available, the vast majority (96.4%) began operations before our study period. Information on when a site ceased operations is missing for all TSDFs in our data, suggesting that these facilities are still in operation. Discussions with staff at the EPA's ORCR suggest that there is little to no variation in TSDF operations over time.

treatment effect equations below. Let d denote the treated group distance bin (e.g., 0–750 m). The triple differences estimate of the percentage change in price due to the opening of a CA is:

$$\mathrm{TE}^{1}_{[d]} = \exp\left(\left[\gamma^{\mathrm{mid}}_{[d]} - \gamma^{\mathrm{pre}}_{[d]}\right] - \left[\gamma^{\mathrm{mid}}_{[\mathrm{ctrl}]} - \gamma^{\mathrm{pre}}_{[\mathrm{ctrl}]}\right]\right) - 1, \tag{2}$$

where $\gamma^{\mathrm{mid}}_{[\mathrm{ctrl}]}$ and $\gamma^{\mathrm{pre}}_{[\mathrm{ctrl}]}$ are the coefficients corresponding to the presence of a TSDF in the respective CA stage in the farther control group distance bin (i.e., 750–1,500 m). Equation (2) illustrates the second and third differencing in our triple differences framework. Our triple differences estimate can be interpreted as a weighted average of the average treatment effect on the treated (ATT).[14] The estimates reflect the average percentage change in price among homes located in distance bin d from a CA, relative to before the CA was initiated. The "1" superscript denotes that this is the first treatment event (i.e., the opening of a CA).

The treatment effect of the second treatment event—completion of cleanup and resolution of the CA—is calculated as:

$$\mathrm{TE}^{2}_{[d]} = \exp\left(\left[\gamma^{\mathrm{post}}_{[d]} - \gamma^{\mathrm{mid}}_{[d]}\right] - \left[\gamma^{\mathrm{post}}_{[\mathrm{ctrl}]} - \gamma^{\mathrm{mid}}_{[\mathrm{ctrl}]}\right]\right) - 1, \tag{3}$$

where $\gamma^{\mathrm{post}}_{[\mathrm{ctrl}]}$ is the coefficient corresponding to the presence of a TSDF in the post-CA stage and located in the control group distance bin (i.e., 750–1,500 m). The pretreatment period in this second quasi-experiment in equation (3) is different from that in equation (2). Here we are interested in how home prices around TSDFs with known contamination issues, and hence an open CA, respond to cleanup and completion of the CA. This comparison and assumed "pretreatment" period is the same as that of Gamper-Rabindran and Timmins (2013) and Haninger et al. (2017), who investigate how home values around sites with already known contamination respond to cleanup-related events.

14. A growing literature is revisiting and questioning the estimand of interest in DID settings where the treatment events are staggered over time (e.g., Goodman-Bacon 2021; Marcus and Sant'Anna 2021; Sun and Abraham 2021). Our identification strategy, however, relies on within-"group" comparisons, where we examine the change in outcomes among a set of treated homes (i.e., those close to a CA site) and a set of control homes that are never treated (i.e., those farther away but still located around the same CA site). For homes around a specific CA site, the treatment event occurs at the same time (i.e., it is not staggered). In this sense, each CA is a separate 2 × 2 DID quasi-experiment. When estimating the regression models we are inherently making this comparison over numerous CAs. To the extent that our tract and county-by-year fixed effects capture broader trends experienced by treated and control homes around each individual CA site, our framework is more akin to a stacked DID study design (Cengiz et al. 2019; Deshpande and Li 2019; Fadlon and Nielsen 2021), which is one of several suggested approaches to address concerns with staggered treatment events (Goodman-Bacon 2021; Roth et al. 2022).

In our more focused DID analysis we drop home transactions that are only near TSDFs where no CA occurs and thus include only homes within 5 km of a CA. Due to the sometimes-clustered nature of TSDFs, however, we still control for the number of nearby TSDFs. Therefore, our spatial DID estimates are also calculated following equations (2) and (3). Although the calculations are the same, dropping the control group of homes near a TSDF where no CA occurs deters a formal triple differences interpretation.

To examine the robustness of our results, in later models we perform coarsened exact matching (CEM) (Blackwell et al. 2009; Iacus et al. 2012)—a pre-regression matching of the data that prunes and re-weights the sample so that the defined treatment and control groups have more balanced distributions in terms of the observed attributes. Homes are matched based on coarsened variables of the number of bathrooms, age, interior square footage, lot acreage, and the percentage of the surrounding area within 500 m that is developed. In addition, we match homes in the treated and control groups only if they are in the same county and sold in the same year. Transactions in the control group are dropped from the matched sample if there are no comparable homes in the treated group in that county and sold in that same year, and vice versa. Details are provided in appendix C.

3.3. Determining the Treatment Group

To determine the spatial extent of the treatment effects, we adopt a strategy similar to Linden and Rockoff (2008), Muehlenbachs et al. (2015), Haninger et al. (2017), and others. Using the sample of $n = 2{,}537{,}669$ homes within 5,000 m of a CA, we estimate a hedonic price regression following equation (1) where $\mathbf{pre}_{ijmt}$, $\mathbf{mid}_{ijmt}$, and $\mathbf{post}_{ijmt}$ are defined using 250 m incremental distance bins from 0–250 m out to 4,500–4,750 m (the farthest distance bin of 4,750–5,000 m is the omitted category). This diagnostic regression includes all other variables in equation (1).

The estimates of $\boldsymbol{\gamma}^{\text{pre}}$, $\boldsymbol{\gamma}^{\text{mid}}$, and $\boldsymbol{\gamma}^{\text{post}}$ are plotted in figure 3, thus displaying the conditional price gradients with respect to distance to a CA in each respective stage. Comparing the pre- versus mid-CA gradients shows that the homes nearest (i.e., within 750 m) to the CA are lesser in value after the CA is opened. Comparison of the post-CA gradient suggests that homes nearest the CA experience a rebound in value back to, and perhaps slightly above, the original pre-CA levels once the CA is complete.[15] The value of homes farther from the sites does not significantly vary across the CA

15. The differences in the point estimates are statistically significant for some, but not all, of the comparisons within 750 m. When comparing the pre- and mid-CA coefficients, Wald tests suggest a statistically significant difference only for the 500–750 m bin ($p = .0221$). Comparison of the mid- and post-CA coefficients suggests significant differences for both the 250–500 m ($p = .0549$) and 500–750 m ($p = .0178$) bins.

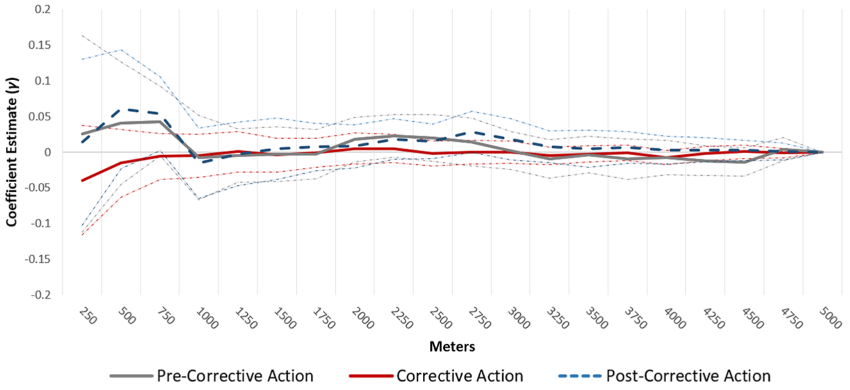

Figure 3. Conditional price gradients with respect to distance from the corrective action site. Light dotted lines display 95% confidence intervals. Distance gradients based on regression coefficient estimates corresponding to 250 m incremental bins. Coefficients estimated from hedonic regression model following equation (1), and using the sample of $n = 2{,}537{,}669$ home transactions within 5,000 m of a corrective action. See section 3.3 for details.

stages. The treated group is thus assumed to be all homes within 0–750 m of a TSDF. The corresponding control group is assumed to be homes within 750–1,500 m.

4. HEDONIC REGRESSION RESULTS

We estimate numerous variants of the hedonic price regression in equation (1). All models include census tract fixed effects, separate county-by-year and county-by-quarter fixed effects, and county-by-year interactions with all house and location attributes unrelated to TSDFs and CAs.[16] The discussion here focuses mainly on the estimated treatment effects expressed in equations (2) and (3), but the full hedonic regression results are presented in appendix E. The control variables that are not of primary interest generally have the expected sign and magnitude and are statistically significant across all models. For example, in early models that constrained $\boldsymbol{\beta}$ to be the same across markets and years, we found that house prices increase with acreage, number of stories, interior square footage, the number of bathrooms, and the percentage of the surrounding land area that is developed. Home values decrease with age and proximity to a highway.

4.1. Price Gradient Associated with Proximity to a TSDF

Following equation (1), our base hedonic specification (model 1) is estimated with the over 9.7 million home transactions within 5 km of at least one TSDF. Before discussing

16. The main results are robust to these modeling decisions and choice of housing bundle attributes. See app. D.1 for details.

how CAs and the related contamination and cleanup activities impact home values, we first examine the general association between house prices and proximity to a TSDF. Using the estimates of $\boldsymbol{\varphi}$, we calculate the price-distance gradient associated with proximity to a TSDF as:

$$\%\Delta p^{s}_{[d]} = \left\{\exp\left(\varphi_{[d]}s\right) - 1\right\} \times 100, \tag{4}$$

where $\varphi_{[d]}$ is an element of the vector $\boldsymbol{\varphi}$ that corresponds to distance bin d (e.g., 0–250 m), and s denotes the number of TSDF sites in distance bin d, which we vary in this illustrative exercise. The term $\%\Delta p^{s}_{[d]}$ is the percentage difference in price associated with a home at a distance d from s TSDFs (e.g., $s = 1, 2, 3$), holding all else constant. These estimates are relative to a constructed counterfactual of a home with no TSDFs within 5,000 m, conditional on all other observed covariates.

The price gradient results in figure 4 demonstrate a strong negative association between proximity to a TSDF and house prices. The top black line suggests that a single TSDF located within 0–250 m of a home corresponds to an 8% decrease in value. This negative association diminishes with distance but remains statistically significant out to 4.75 km. Although measuring distance using incremental 250 m bins allows for a flexible distance gradient, the results align with expectations, suggesting a smooth price gradient that diminishes with distance.

Figure 4 also suggests large cumulative effects, particularly in the nearest distance bins. The dark gray middle gradient suggests that having two TSDFs within 0–250 m is associated with a 15% depreciation in home values, and the presence of three TSDFs is associated with a 22% decline. These negative associations also diminish with distance, but the price decrements again remain significant out to 4.75 km. Although

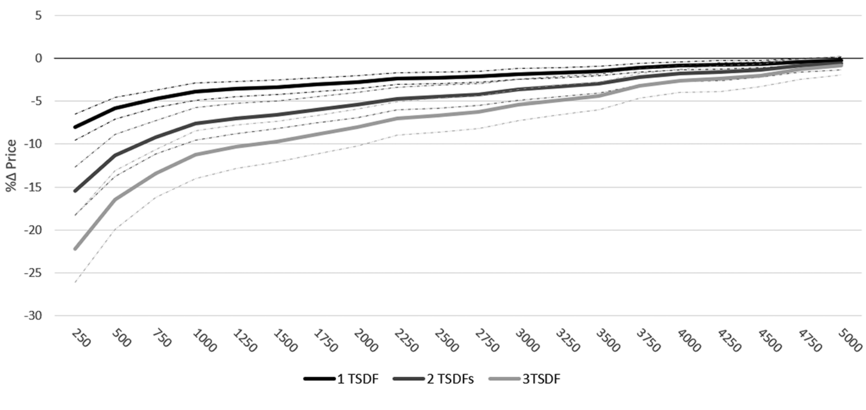

Figure 4. Estimated price gradients with respect to proximity to TSDFs (model 1). Based on estimates from model 1 (see app. E for full regression results). The 95% confidence intervals are displayed by dotted lines.

we cannot argue these estimates as causal, it is clear that homes near a TSDF already experience a significant discount in value, irrespective of any contamination issues.

4.2. Price Effects of Corrective Actions

The remainder of the analysis aims to estimate the causal effects of a CA on these already value-depressed homes. Model 1 in table 3 is estimated using all transactions of homes within 5 km of a TSDF. The triple differences estimate from model 1 suggests a 5.5% decrease in value, on average, when a CA is initiated at a TSDF within 750 m. When the contamination issue is resolved and the CA complete, there is an average appreciation of 7.2% among homes within 750 m.

Model 2 in table 3 is similar to the previous model but is estimated using only transactions of homes that are within 5 km of a TSDF where a CA has occurred as of the time of the sale or will occur later in the study period. TSDF sites where CAs occur, and the homes and neighborhoods around those sites, may be systematically different from those around TSDFs where a CA does not occur. Focusing on the former, this more homogeneous sample in model 2 facilitates a cleaner DID comparison. The results are quite similar to the previous model, suggesting a 5.3% decline in value when a CA is opened and a subsequent 6.6% rebound in price once the CA is complete.[17]

To provide an even cleaner DID quasi-experiment, we next estimate the model using only homes in the treatment (0–750 m) and control (750–1,500 m) distance bins. In other words, we exclude all home sales that are not within 1,500 m of a CA site. In doing so, model 3 in table 3 suggests a smaller and insignificant 2.8% depreciation from the opening of a CA. The estimated 6.3% increase in value after completion of a CA remains robust.

Finally, to further examine the robustness of our results, we parse and re-weight this 0–1,500 m sample using CEM techniques (Blackwell et al. 2009; Iacus et al. 2012). The matching procedure substantially reduces the sample size but also results in more comparable treatment and control groups.[18] Model 4 in table 3 shows that the opening of a CA is again statistically insignificant. However, the average price impact from the cleanup and resolution of the contamination issue remains robust, suggesting a 7.4% increase in the value of homes within 750 m.

We also explored the possibility of including house-level fixed effects. The house fixed effects variants of the models yield smaller and insignificant results (see table A9; tables A1–A18 are available online). We cannot unambiguously claim what is driving the sensitivity of our results to the inclusion of house fixed effects. It could be that the house fixed effects validly control for remaining unobserved, time-invariant and

17. Similar results are found when allowing for treatment effect heterogeneity for each 250 m incremental bin out to 750 m. Wald tests support our decision to allow for a single 0–750 m treatment bin. See app. D.2 for details.

18. See sec. 4.3 in the main text and app. F.1 for details.

Table 3. Percentage Change in Price for 0–750-Meter Bin

	Model 1 TSDF 5 km Sample Triple Differences	Model 2 CA 5 km Sample DID	Model 3 CA 1,500 m Sample DID	Model 4 CA 1,500 m CEM Sample DID
Corrective action opened				
0–750 m	−5.4679**	−5.2688**	−2.8365	−2.4289
	(2.1754)	(2.1969)	(2.1730)	(2.3581)
Corrective action completed				
0–750 m	7.2119***	6.6418***	6.2789**	7.4067***
	(2.4706)	(2.4245)	(2.5713)	(2.6732)
Tract fixed effects	Yes	Yes	Yes	Yes
House and location attributes	County × Year	County × Year	County × Year	County × Year
Year fixed effects	County × Year	County × Year	County × Year	County × Year
Quarter fixed effects	County × Quarter	County × Quarter	County × Quarter	County × Quarter
Observations	9,762,316	2,537,669	203,550	100,507
Adjusted R-squared	.784	.784	.789	.811

Note. Standard errors in parentheses, clustered at the county level. Estimates calculated following eqs. (2) and (3) using the "nlcom" command in Stata 17/MP and are based on the coefficient estimates from the hedonic regression models 1–4. Full regression coefficient results are presented in app. E.

* $p < .10$.

** $p < .05$.

*** $p < .01$.

property-specific confounders, but we also posit that the dramatically reduced number of identifying observations due to the inclusion of house fixed effects is a key driver behind the sensitivity of the results. For example, the number of posttreatment identifying observations when estimating the price effects associated with the opening of a CA decreases from 28,312 transactions within 0–750 m of a CA site in the full sample to just 829 sales in the house fixed effects sample—a 97% decrease. This drop is largely attributed to there being few repeat sales where the same home is sold before and after the opening of a CA. Such transactions are necessary because identification in a house fixed effects model relies solely on within-house variation. Similarly, the number of posttreatment transactions for completion of CA decreases from 3,459 in the full sample to just 563 in the house fixed effects sample—an 84% decrease in the number of identifying observations.[19]

Additional analysis suggests that it is not just the inclusion of house fixed effects that is driving the sensitivity of the results but that changes in the sample of identifying observations play a critical role. The transactions used for statistical identification in a house fixed effects model are fewer, less representative, and systematically different from the rest of the sample. For example, in the original models the 3,459 post-CA completion sales within 0–750 m cover 81 counties in 27 different states. When focusing on the corresponding 563 identifying sales in the house fixed effects model, this is reduced to just 34 counties in 14 states. Transactions used for statistical identification in the house fixed effects models also tend to be of homes lower in value, smaller, and in more urban areas (see app. B.3 for details). It is possible that less wealthy (perhaps less educated) households living in these homes are less informed and responsive to changes in environmental quality. Further details and analysis are provided in appendixes B.3 and D.3.

Despite the sensitivity of our results to the inclusion of house-level fixed effects, this study is still useful in informing policy. Our analysis covers virtually the entire universe of TSDFs across the contiguous United States and is conducted using a sample that, to the largest extent possible, includes all transactions of single-family homes. Despite the large overall sample of over 9.7 million transactions, the number of identifying observations is relatively small, particularly for models attempting to include house fixed effects. The estimated price effects of CA completion are otherwise robust across modeling and sample decisions and are consistent with findings in the literature (Gamper-Rabindran and Timmins 2013; Haninger et al. 2017; Guignet et al. 2018; Cassidy et al. 2022).

4.3. Assessing a Causal Interpretation

A causal interpretation of the estimated treatment effects from the initiation and completion of a CA depends on the validity of the assumed counterfactual. To assess the

19. Further details are provided in app. B.3, and in particular, table A4.

appropriateness of a causal interpretation we first compare the mean values of observed covariates across the treated and control groups (i.e., homes within 0–750 m and 750–1,500 m of a CA, respectively). We do this for the full sample of homes in these groups and then for the CEM-weighted sample. Second, we compare the univariate distributions of observed characteristics using the *L1* statistics (Blackwell et al. 2009; Iacus et al. 2012). Third, we assess the comparability of the types of home being sold before and after each of the two treatment events. And fourth, we evaluate the validity of the common trends assumption (Angrist and Pischke 2009) by conducting an event study where we estimate models that include lead and lag terms around the CA opening and completion events. The pretreatment trends are then compared across the treated and control groups.

A series of two-sample *t*-tests comparing the full, unweighted samples of transactions within 0–750 m and 750–1,500 m of a CA reveals statistically significant differences in the mean values for all covariates (see table A12). However, these sample sizes are fairly large, resulting in small standard errors. The differences in means are not economically significant in most cases. For example, the average acres are 0.245 versus 0.237 acres, the average number of bathrooms is 1.74 versus 1.72, and the difference in interior square footage is only 62 square feet. The differences in means of the location variables are slightly starker; for example, 50% of the land area within a 500 m buffer of a home in the control group is developed, on average, compared to 43% for the average treated home. Similarly, 47% of the control group is within 500 m of a major highway, compared to 58% of the treated group. The average number of TSDFs in close proximity is similar across the two groups (1.21 vs. 1.23 TSDFs within 1,500 m), but again this difference is statistically significant.

As expected, the mean values among the CEM weighted sample are more comparable across the treated and control groups, with statistically significant differences ($p \leq .10$) only among six of the 10 covariates. And even when there are statistically significant differences, there is again little economically meaningful difference. The percentage difference across all variable means is less than 3%, with the exception of proximity to a highway (50% of the control group is within 500 m of a highway, compared to 58% of the treated group—a 15% difference).

Comparison of the multivariate and univariate *L1* statistics (Blackwell et al. 2009; Iacus et al. 2012) shows that the univariate and multivariate distributions across the treated and control groups are more similar after matching (see tables A13, A14). This is true even among most of the covariates that were not included in the CEM algorithm.[20] It is worth emphasizing that we restrict matches to only homes in the same county and that sold in the same year (in addition to matching on the coarsened variables discussed in sec. 3.2). Doing so draws focus to CA sites where both treated

20. The few exceptions are that there was a slight increase in the univariate *L1* statistics for housing age, an indicator denoting that the number of stories is missing, and the number of TSDFs within 0–5,000 m.

and control transactions are observed and, in general, helps mitigate biases associated with a particular county and year by ensuring an equal balance of treated and control observations in each county-year combination.

Next, focusing on home transactions in the 0–750 m treated group bin, we compare whether the types of homes sold before and after the opening of a CA, as well as the completion of a CA, are systematically different. Two-sample *t*-tests show that the differences in mean values of observed covariates are often statistically, and perhaps also economically, significant (see table A15). We control for these observed attributes in the hedonic models, but this could suggest systematic differences in unobserved characteristics as well, which supports the inclusion of the numerous spatial and spatiotemporal fixed effects in our models.

To provide a more thorough assessment that better accounts for those spatiotemporal fixed effects, we estimate a series of linear probability models of posttreatment status for both the opening and completion of a CA. The house and location attribute coefficients are largely insignificant, both individually and jointly (see app. F.2 for details). Conditional on our spatial (tract) and spatiotemporal (county-by-year and county-by-quarter) fixed effects, the regression results suggest no systematic differences in the types of homes sold before and after each treatment event, thus lending credibility to a causal interpretation of the estimated price effects from a CA.

Finally, in a well-defined quasi-experiment the trajectory of the outcome variable experienced by the treated group in the absence of treatment must be the same as that of the assumed control group in the posttreatment period. This is commonly referred to as the common or parallel trends assumption (Angrist and Pischke 2009). Given that we do not observe the true counterfactual (i.e., the treated group absent the treatment), researchers often turn to the pretreatment trends. If the outcome of interest (i.e., house prices) for the treated and control groups follow similar trends in the pretreatment period (i.e., before the opening or completion of a CA), then it is more reasonable to assume that those trajectories would have remained similar in the absence of the treatment.

We conduct an event study where we reestimate a variant of model 2 that includes lead and lag variables for both the opening and completion of a CA, grouped into five-year intervals. Figure 5 displays the differences in trends across the treated (0–750 m) and control (750–1,500 m) groups.[21] Panel *A* shows the event study results around the opening of a CA, and panel *B* shows the same for the completion of a CA. The event study graphs for both treatment events are estimated from the same regression model, but the assumed "pretreatment" periods differ. In panel *A*, the lead estimates reflect the difference in the pre-CA trends, but in panel *B* the differences in the lead coefficients represent the mid-CA period. The latter pretreatment period differs because, similar to analyses by Gamper-Rabindran and Timmins (2013), Haninger et al.

21. The undifferenced trends for the treated and control group are displayed in app. F.3.

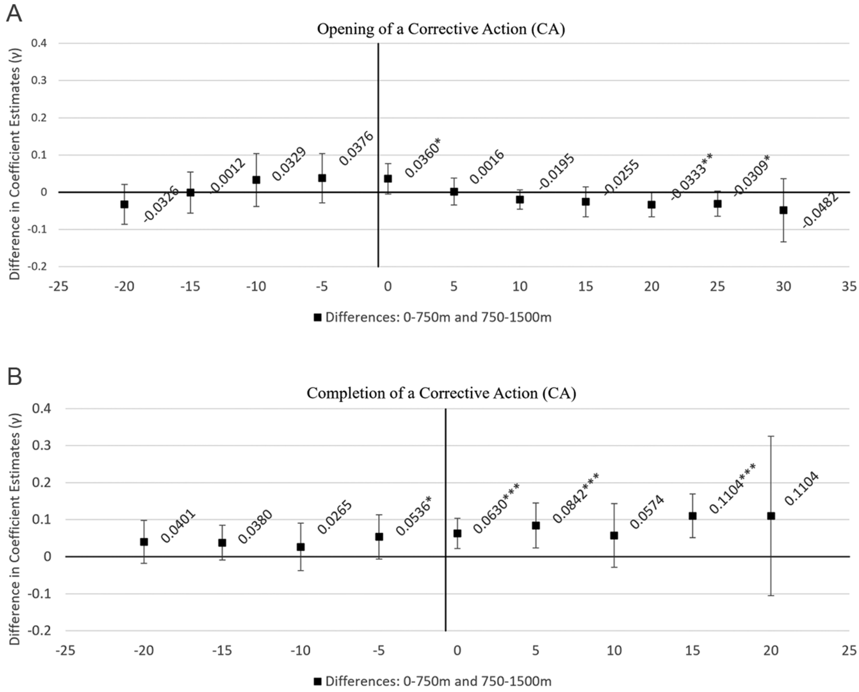

Figure 5. Event study of opening and completion of a corrective action: differences in trends. The estimates are based on a variant of model 2 where lead and lag coefficients are grouped into five-year intervals. Panel *A* shows the event study results around the opening of a corrective action (CA), where the lead estimates reflect the difference in the pre-CA trends between the treated (0–750 m) and the control (750–1,500 m) groups. The lag coefficients in panel *A* reflect those same differences during the mid-CA period. Panel *B* shows the differences in the trends before and after the completion of a CA. In panel *B* the differences in the lead coefficients represent the mid-CA period, and the lags correspond to the post-CA period. Event study graphs for both treatment events are estimated from the same regression model, but the assumed "pretreatment" period differs. Note that the x-axes are scaled differently to reflect the observed range in pre- and posttreatment years. The undifferenced trends are shown in apprndix F.3, along with additional details and robustness checks. $^{*}p < .10$; $^{**}p < .05$; $^{***}p < .01$.

(2017), and others, we are interested in identifying how cleanup completion affects home values compared to a baseline where contamination is present and known.

First, consider the event study results pertaining to the opening of a CA. The differences in the lead coefficients shown in panel *A* are not statistically different from zero. But even if there were significant differences, the parallel trends assumption merely requires that those differences be constant over the pretreatment period. Wald tests fail to reject the null hypothesis that the differences are statistically equal across the lead periods ($p = .1146$). This is consistent with the parallel trends assumption but only marginally so, and the point estimates suggest that the pre-trends are not parallel. The pre-trends even cross between 10 and 15 years prior to the opening of a CA, as can be seen by the shift in the differenced point estimates from negative to positive in figure 5.

There are always inherent, irreducible lingering threats to causal inference in observational settings such as ours. Nonetheless, the event study results around the completion of a CA, as shown in panel *B* of figure 5, are in line with a causal interpretation. Visual inspection of the point estimates for the differences in the lead coefficients suggests that the pre-trends are parallel. Wald tests provide further support. We fail to reject the null hypothesis that the differences in the lead coefficients between the treated and control groups are statistically equal across the lead periods ($p = .7344$). One could question the slight marginally significant increase in price among the treated group within five years before CA completion, but we emphasize that the completion date is a proxy for when cleanup is complete and residents perceive a site as safe. Given the often long duration of cleanup activities at these sites, it is reasonable that any price appreciation due to cleanup may start to be experienced earlier—when residents can see the remediation efforts, and intermediate milestones are achieved. In any case, we emphasize that this marginally significant difference is not statistically different from the earlier lead differences, which is again consistent with the parallel trends assumption. We find similar results when estimating these event study graphs with other models and samples (see app. F.3).

5. WELFARE IMPLICATIONS

We find strong evidence that CA completion leads to an increase in the value of homes within 0–750 m and therefore focus the welfare discussion here on CA completion. We reestimate a variant of model 2 but focus on CA completion by dropping the 16,268 transactions within 0–750 m or 750–1,500 m of a pre-CA TSDF. The corresponding pre-CA variables and coefficients in equation (1) are no longer identified and are also dropped from the hedonic price models estimated here. This allows for a simpler DID model that is focused solely on CA completion. As seen by model 5 in table 4, the results are similar, suggesting that CA completion leads to a 5.9% appreciation to homes within 750 m.

The data suggest that there are 25,415 single-family homes (not just transactions) within 750 m of one of the 195 TSDFs where the CA was fully complete by 2018. The mean transaction price for the corresponding 627 sales of these homes in 2018 is about \$206,950. We calculate the mean capitalization effect of cleanup as: $206{,}950[\mathrm{TE}^2_{[0-750]}/(1 + \mathrm{TE}^2_{[0-750]})]$, where $\mathrm{TE}^2_{[0-750]}$ is the percentage change in price expressed by equation (3). Based on the 5.9% estimated appreciation from model 5 we calculate a mean capitalization effect of \$11,605 per home. Multiplying this average by the 25,415 single-family homes suggests that the completion of cleanup at these 195

Table 4. Policy Illustration of Corrective Action Completion: Capitalization Effects versus Ex Post Lower Bound Welfare Estimates

	Model 5 Time Invariant CA Effects	Model 5′ Time Variant CA Effects: Year Indicators	Model 5″ Time Variant CA Effects: Linear Trend
Assumed ex post price surface ($\tilde{t}$)	NA	2018	2018
Percentage increase	5.9408***	3.0743	7.2590***
	[1.4944 - 10.3872]	[−2.9525 - 9.1011]	[1.8282 - 12.6898]
Mean price increase (2018$)	$11,605***	$6,172	$14,006***
	[3,406 - 19,804]	[−5,567 - 17,912]	[4,237 - 23,775]
Total increase (millions 2018$)	$294.9***	$156.9	$356.0***
	[86.6 - 503.3]	[−141.5 - 455.2]	[107.7 - 604.2]

Note. The 95% confidence intervals are displayed in brackets. The first column (model 5) displays the estimated capitalization effects based on a variant of model 2 that is estimated from a sample that excludes all sales with a pre–corrective action TSDF within the treatment or control group distance bins (0–750 m or 750–1,500 m, respectively). The sample size is 2,521,404 transactions. The second column (model 5′) displays the estimated price changes along a fixed 2018 price surface and is based on a variant of model 5 where all coefficients with respect to the CA stages are interacted with binary indicators for each year. The third column (model 5″) also displays the estimated price changes along a fixed 2018 price surface but is based on a variant of model 5 where all coefficients with respect to the CA stages are interacted with a linear time trend variable, which ranges from 0 (= 2000) to 18 (= 2018). Models 5′ and 5″ include separate interaction terms to allow the coefficients corresponding to the presence and number of TSDFs in each incremental 250 m bin to vary by year and by county. The full hedonic regression results are reported in table A17.

* $p < .10$.

** $p < .05$.

*** $p < .01$.

TSDFs increased total housing stock value by $295 million. If similar capitalization effects are expected upon completion of cleanup at the remaining 406 TSDFs with an open CA as of the end of 2018, then we would see the value of the 41,973 homes within 750 m of those sites increase by a total of $487 million.

Estimating the capitalization effects of hazardous waste cleanup as done above is intuitively appealing, but these estimates generally lack a formal welfare interpretation if the actual hedonic price surface is changing over time (Klaiber and Smith 2013; Kuminoff and Pope 2014). We next pursue a formally valid, lower-bound ex post estimate of the benefits of cleanup completion. Banzhaf (2021) demonstrates that a change in price along the same ex post price gradient is a lower bound of the Hicksian equivalent surplus for an improvement in quality. To carry out the approach the entire estimated hedonic price surface must be allowed to vary over time; in our context this includes the slope coefficients with respect to the TSDF and CA variables of interest. We adapt the hedonic price model in equation (1) by adding interaction terms to allow the TSDF and CA coefficient vectors $\boldsymbol{\varphi}_t$, γ_t^{mid}, and γ_t^{post} to vary over time. Details are provided in appendix G.

An additional complication in a nationwide context such as ours is that the price surface would ideally be allowed to also vary spatially across markets. From a theoretical standpoint, the ideal model might include interaction terms between the TSDF and CA vectors with county-by-year indicators. However, given the highly localized nature of this disamenity, the resulting distance-by-county-by-year bins would have very few and often zero identifying observations. Instead, the hedonic model proposed here allows for heterogeneity by including separate interactions by market and by year with $\mathbf{TSDF}_{ijm}$. The coefficients corresponding to the mid- and post-CA variables are allowed to vary over time but not across geographic markets.[22]

Given the empirical constraints in the current application, this is the best we can do in applying Banzhaf's (2021) framework to a national setting. The most necessary feature for bounding welfare calculations is maintained—the price gradients with respect to CAs are allowed to vary over time. We first do this by interacting $\mathbf{mid}_{ijmt}$ and $\mathbf{post}_{ijmt}$ with binary year indicators, but due to empirical concerns described below, we also estimate this model by interacting $\mathbf{mid}_{ijmt}$ and $\mathbf{post}_{ijmt}$ with a linear time trend (see app. G for details).

Similar to equation (3), the coefficient estimates from these models are used to estimate the post-CA treatment effects specific to the ex post hedonic price surface in year $\tilde{t}$. For that same assumed ex post year $\tilde{t}$, we then calculate the mean price increase resulting from CA completion as well as a total increase in value across all impacted homes. These estimates represent a theoretical lower bound of the Hicksian equivalent surplus for an improvement in quality (Banzhaff 2021).

A key question in the current context and in other policy settings is, What is the appropriate ex post hedonic price gradient to assume for calculating changes in welfare? We find that the estimates are quite sensitive to what year is assumed for $\tilde{t}$. As shown in figure 6, the resulting estimates vary from year to year, ranging from an insignificant −8% to a positive and significant 20%. The price effects in model 5′ in figure 6 are allowed to vary freely over time, but the estimates fluctuate greatly from year to year, are not very precise, and have wide confidence intervals. We attribute this largely to the small number of identifying post-CA transactions observed in any given year; especially in the earlier years where we see the estimates vary more widely (see fig. A8).

A more restrictive model can be estimated that would still allow the CA price effects to vary over time but would "smooth-out" year-to-year fluctuations. A trend line is

22. As a robustness check, we do investigate spatial heterogeneity of the CA price effects based on broader regional market definitions and find statistically insignificant differences across markets (see app. G for details).

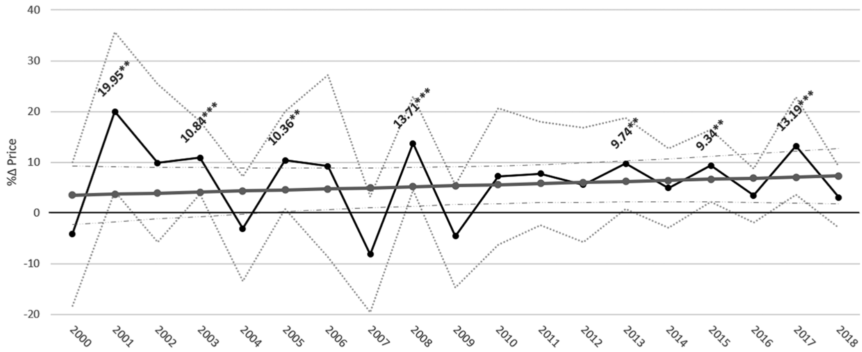

Figure 6. Price impacts of corrective action completion by year. The solid black line shows the estimates of the percentage changes in price due to completion of a CA, based on the results from model 5′ and following equation (A2) in appendix G. Labels are included for only the statistically significant estimates ($p < .10$). The solid gray line shows how the percentage change in price estimates varies by year when a linear trend is assumed, based on model 5″ and following equation (A6) in appendix G. The 95% confidence intervals are displayed by the dashed or dotted lines. The point estimates from model 5″ are not labeled but are statistically significant from 2003 and onward. *$p < .10$; **$p < .05$; ***$p < .01$.

fitted in figure 6 by reestimating the hedonic price model (model 5″) and allowing the CA coefficients to vary only linearly over time. This allows the hedonic price surface with respect to CA completion to vary, albeit in a more restrictive fashion. The coefficients on the linear trend interaction terms are jointly insignificant ($p = .3533$), suggesting that the price effects of CA completion do not significantly change over time (i.e., the linear trend from 2000 through 2018 is flat). Under the assumption of a constant CA completion price effect over time, our earlier estimates of the capitalization effects can be interpreted as a lower-bound estimate of the ex post benefits of cleanup.

If, however, we want to allow the price effects to shift over time following our linear trend assumption, then we can do so based on the fitted line in figure 6. The estimates start with a statistically insignificant 3.5% appreciation in 2000 and then gradually increase over time, first suggesting a statistically significant positive effect in 2003 of 4.1% and then continuing to increase up to 7.3% in 2018.

In order to apply Banzhaf's (2021) framework and focus on the ex post price surface, we use the estimates from model 5″ and assume $\tilde{t} = 2018$ for the welfare calculations. As shown in table 4, doing so suggests a significant 7.3% appreciation from the completion of cleanup. This translates to an average lower-bound ex post benefit of about \$14,000 per household living within 750 m. The lower-bound benefit from all 195 TSDFs remediated through the corrective action program since its inception is estimated to be \$356 million.

If we instead use the estimates from model 5′, then based on the 2018 price surface we estimate benefits that are statistically insignificant and more than half in magnitude. As described above, however, this largely depends on the assumed year for the ex post price surface. Although 2018 makes sense for this illustration, other ex post years could be assumed, and as shown by figure 6 the estimated benefits are sensitive to this assumption. For example, assuming an ex post year of 2017 yields a statistically significant estimate of $613 million.

We prefer model 5″ for the welfare calculations because it allows for temporal variation in the hedonic price surface but also smooths out noise in the estimates from any given year. In other applications where data limitations may deter estimation of the ideal, most flexible specifications, researchers may want to consider imposing more structure on the assumed functional form when applying Banzhaf's (2021) framework for bounding welfare changes and discuss the trade-offs.

6. CONCLUSION

There are 2,389 treatment, storage, and disposal facilities (TSDFs) of hazardous waste in the contiguous United States, and over 487 open corrective actions with ongoing investigations and incomplete cleanups at these sites. Considering RCRA sites more broadly, the US EPA (2013) estimates that over 35 million people (12% of the US population) live within one mile of a corrective action (CA). Releases and improper disposal and treatment of hazardous chemicals clearly remain a threat to local communities. At the same time, monetized benefits of cleanups to nearby residents have generally not been accounted for in benefit-cost analyses informing federal regulations under RCRA. To fill this gap we framed two sets of quasi-experimental comparisons using a novel, high-resolution data set that combines nationwide data on housing transactions and TSDFs. We find that CAs do affect home values up to 750 m away. This raises potential environmental justice concerns because we also find that home values in communities around TSDFs are already significantly depressed, irrespective of any contamination or cleanup activities.

Our results suggest that the discovery of contamination and opening of a CA investigation led to a price decline of up to 5% among homes within 750 m. This result is not robust across all models, however, and potential differences in the pretreatment trends suggest that causal inference may not be appropriate. These findings are not necessarily surprising. The opening of a CA may not signal new information that changes local residents' and potential homebuyers' risk perceptions. Any perceived risks may already be capitalized in surrounding home values due to visual cues associated with proximity to a TSDF in general. In fact, often a new CA involves investigation of historical contamination issues (US EPA 2013) and not necessarily a new release.

We do find strong, causal evidence that the completion of a CA leads to a 6%–7% increase in value among homes within 750 m. The completion of a CA signals resolution of the contamination issue, either through the completion of active cleanup efforts, placement of physical barriers to prevent exposure and pollutant migration, and/or future land use restrictions to minimize exposure. These results are robust to a variety of approaches to control for spatially and temporally correlated factors that could otherwise confound the analysis, including implementation of difference-in-differences (DID) and triple differences methods, the use of census tract fixed effects, allowing for county-specific year and quarter fixed effects, and through the use of coarsened exact matching (CEM). Examination of the pretreatment trends further supports causal inference.

Despite the sensitivity of our results to the inclusion of house fixed effects, this study is useful in informing decision-makers. Our analysis covers virtually the entire universe of TSDFs across the contiguous United States and is conducted using a sample that, to the largest extent possible, includes all available transactions of single-family homes. Despite the large overall sample of over 9.7 million transactions, we find that the number of identifying observations is much smaller and systematically different when including house fixed effects. Investigation of the selection versus omitted variable bias trade-off associated with house fixed effects or repeat sales models in general is a fruitful direction for future research.

Our estimated 6%–7% post-cleanup appreciation among homes within 750 m is otherwise robust across modeling and sample decisions, and is consistent with findings in the literature. Hedonic property value studies of other cleanup programs under EPA's Office of Land and Emergency Management suggest that cleanup leads to a 5%–20% increase in surrounding home values (Gamper-Rabindran and Timmins 2013; Haninger et al. 2017; Guignet et al. 2018), although those studies find farther extending effects, out to 2–4 km. Our results are also in agreement with Cassidy et al.'s (2022) census tract–level analysis of the CA program. They find that cleanup leads to a roughly 11% appreciation among homes in the census tract where the RCRA site is located. Cassidy et al. also find these price effects to be the largest among homes in the lower deciles of the tract-specific price distributions, which as suggested by our study, likely correspond to the homes nearest the RCRA sites.

Our treatment effect estimates can be used to inform policy. Model 5 suggests a 5.9% appreciation upon completion of a CA, which translates to a mean capitalization effect of about $11,600 per home. Considering the 25,415 single-family homes within 750 m of one of 195 TSDFs where a CA was complete by 2018 (the end of our study period), this suggests that cleanup of these sites led to a total increase in housing stock value of $295 million.

Following a similar exercise, we apply an approach proposed by Banzhaf (2021) to estimate a lower bound of the ex post benefits of cleanup completion. The results are sensitive to the assumed ex post year and specification in allowing for temporal variation in the hedonic price surface, but under our preferred model we find an average lower-bound benefit estimate of about $14,000 per household. This implies a total lower-bound benefit of $356 million from the 195 TSDFs that have been cleaned up since the inception of the CA program. These benefits likely reflect changes in perceived and actual health risks, improvements in local environmental quality, and possibly reductions in other displeasing aesthetics.

The estimated benefit of cleanup at a single TSDF will vary greatly depending on the number of nearby residents, among other things, but our estimates suggest an average lower-bound benefit of $1.83 million per TSDF (= $356 million/195 TSDFs). The costs of cleanup also vary substantially but in general would tend to outweigh our average lower-bound estimate of the benefits.[23] Nonetheless, we emphasize that the estimated benefits are a theoretical lower bound (Banzhaf 2021), and our estimates only account for households living in single-family homes near these sites. Populations living in other types of housing, which are often more prevalent in urban areas, are not included. Additionally, our estimated benefits to nearby residents are just one part of the benefit calculus; there are other endpoints and impacted populations to consider.[24]

Our estimated price and welfare impacts reflect the average effects of the opening and completion of a CA, but TSDFs are local disamenities, and there is likely significant variation in the impacts around any one site. TSDFs can differ in baseline characteristics, community awareness, and actual and perceived risks. The CA events can also vary in terms of contamination severity, presence of exposure pathways, and the extent and duration of subsequent cleanup activities. Examination of such heterogeneity is a useful direction for future research and could yield important implications for benefit transfer and policy analysis.

Hedonic property value methods compose an increasingly large portion of the nonmarket valuation literature but are seldom used in policy analysis (Petrolia et al. 2021). One deterrent is the lack of a formal welfare interpretation in many contexts. Theoretical work building on that of Kuminoff and Pope (2014), Banzhaf (2020, 2021), and others to provide more formal interpretations, and ideally in both ex post and ex ante settings, will help further the use of hedonic property value results in informing policy. Nonetheless, our average capitalization effects and lower-bound benefit estimates serve as a useful intermediate step to inform policy on the cleanup of hazardous waste sites.

23. Case studies described by EPA (2013) suggest costs ranging from $2.7 million to $1.9 billion. Tonn et al. (1991) simulated remediation costs across 2,000 different scenarios and found a base case average cost of $12.0 million per site. The BCA for the EPA's Hazardous Waste Generator Rule mentions a median remediation cost of $7.1 million (US EPA 2016). The 2014 Definition of Solid Waste BCA notes an average cleanup cost of $8.7 million and a median cost of $2.5 million among recyclers of hazardous material (US EPA 2014). Cost estimates are adjusted to 2019$ using the Bureau of Labor Statistics's annual average consumer price index for all urban consumers (https://www.bls.gov/cpi/tables/supplemental-files/historical-cpi-u-202105.pdf; accessed June 14, 2021).

24. A brief, high-level review of benefit categories is provided by US EPA (2013). For more detailed discussions, see the regulatory impact analyses cited in table A1.

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